Reg.No :.....

Name :.....

MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS)
SECOND SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG2DSCECT101-DATACOMMUNICATION

Duration:1.5hoursMaximumMarks:50

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

PartA

MultipleChoiceQuestions

Answer **All** Questions

Eachquestioncarries**1**mark

Write complete answers of your choice in the answer paper against each question number

1. Identifythetransmissionmodethatallowsdatatoflowinbothdirectionsbutonlyonedirectionata time.[U][1]

a) Simplex b)HalfDuplex c)FullDuplex d)Multiplexing

2.Whatunitisusedtomeasurethebandwidthofasignal? [U] [1]

a)Bitrate b)Baudrate c)Hertz d)Decibel

3.Whatistheprimaryfunctionofarepeaterinadatacommunicationsystem? [U] [1]

a)Amplifyingdatasignals

b)Encodingdata

c)Multiplexingdata

d)Filteringdata

4. Thermalnoiseisintroducedbywhichfactorinacommunicationsystem? [U] [1]

a)Impulsenoise b) Crosstalk c)Thermal noise d)Intermodulationnoise

5. Amongguidedmedia,whichoneprovidesthehighestdatatransmissionrate? [U] [2]

a) STPb)UTP c)Coaxialcable d)Fiberopticcable

6. Identifythewirelesstransmissionmediumthatreliesonline-of-sightcommunication[U] [2]

a) Infrared b)Radiowavec)Satellitemicrowaved)Fiberopticcable

7.HowdoesAmplitudeShiftKeying(ASK)representdigitaldata? [U] [3]

a)Byvaryingtheamplitudeofthecarriersignal

b)Bychangingthefrequencyofthecarriersignal

c)Bymodifyingthephaseofthecarriersignal

d)Bycombiningamplitudeandphasechanges

8.WhatprincipledoesPhaseShiftKeying(PSK)relyonfordatatransmission? [U] [3]

a)Amplitudevariation

b)Frequencymodulation

c)Phasechanges

d)Timedivision

9.Choosethemultiplexingtechniquecommonlyusedinfiberopticcommunication?

a)FDM b)TDM c)WDM d)FHSS [U] [3]

10.Themodulationtechniquepreferredforlong-distanceanalogsignaltransmissionis [U] [3]

a) AmplitudeModulation

b) FrequencyModulation

c) PhaseModulation

d) DeltaModulation

(1 x 10 = 10)

PartB

ShortAnswerQuestions

Answer Any 4 Questions

Each question carries **5** marks

11. Withthehelpofablockdiagramexplainthefivecomponentsinvolvedinacommunicationsystem.[U][1]
12. Distinguishbetweenhalf-duplexandfull-duplextransmissionmodes? [U] [1]
13. Describethestructureofacoaxialcableanditsapplicationsindatacommunication.[U][2]
14. Howdoesskypropagationdifferfromline-of-sightpropagation? [U] [2]
15. Comparesynchronousandasynchronoustransmission . [U] [3]
16. Explainfrequencyshiftkeying [U] [3]

(4 x 5 = 20)

PartC

EssayQuestions

AnswerAny **2**Questions

Each question carries **10** marks

17. Explainthedifferenttypesoftransmissionimpairmentsindatacommunicationandtheireffects. [U] [1]
18. ComparePulseCodeModulationandDeltaModulation[U][2]
19. DescribetheworkingofFrequencyHoppingSpreadSpectrum(FHSS)andDirectSequenceSpread Spectrum (DSSS) techniques with their advantages.[U][3]

(2 x 10 =20)